Calculus and Linear Algebra

Exponential, Logarithmic and Trigonometric Derivatives

Lecture 19

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28 October 2026

Lecture outline

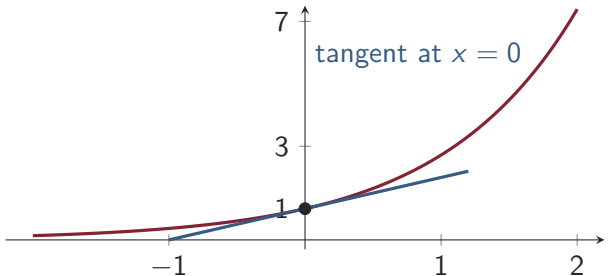
- Exponential functions: e^x and a^x.
- Logarithmic functions: $\ln x$ and $\log_a x$.
- Trigonometric derivatives: $\sin x$, $\cos x$, $\tan x$.
- Chain rule with elementary functions.
- Logarithmic differentiation.
- General powers and the example x^x.
- Higher-order derivatives.
- Mixed exercises.

Derivative of the exponential function

The exponential function with base e has the remarkable property

$$\boxed{\frac{d}{dx}e^x = e^x.}$$

Its derivative is equal to the function itself.



Exponential functions with another base

For $a > 0$, $a \neq 1$,

$$\boxed{\frac{d}{dx} a^x = a^x \ln a.}$$

Indeed,

$$a^x = e^{x \ln a}.$$

Applying the chain rule:

$$\frac{d}{dx} e^{x \ln a} = e^{x \ln a} \ln a = a^x \ln a.$$

Special case

For $a = e$, $\ln e = 1$, so we recover $(e^x)' = e^x$.

Chain rule with exponentials

If $u = g(x)$, then

$$\boxed{\frac{d}{dx} e^{g(x)} = e^{g(x)} g'(x).}$$

Examples:

$$\frac{d}{dx} e^{2x} = 2e^{2x},$$

$$\frac{d}{dx} e^{x^2} = 2xe^{x^2},$$

$$\frac{d}{dx} e^{1/x} = -\frac{1}{x^2} e^{1/x}, \qquad x \neq 0.$$

Example: product with an exponential

Differentiate

$$f(x) = x^2 e^x.$$

Use the product rule:

$$f'(x) = 2xe^x + x^2 e^x.$$

Factor:

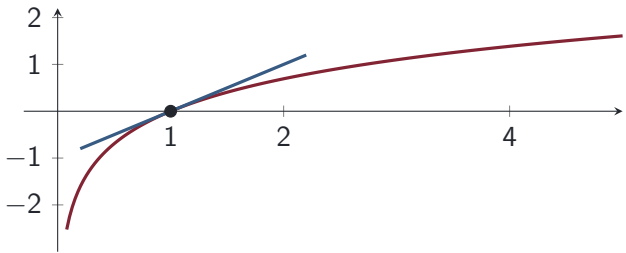
$$\boxed{f'(x) = xe^x(x+2).}$$

The exponential factor is always positive, a fact that will later simplify sign analysis of f'.

Derivative of the natural logarithm

For $x > 0$,

$$\boxed{\frac{d}{dx} \ln x = \frac{1}{x}.}$$



At $x = 1$, the tangent slope is 1.

Logarithm with a general base

For $a > 0$, $a \neq 1$,

$$\log_a x = \frac{\ln x}{\ln a}.$$

Therefore

$$\boxed{\frac{d}{dx} \log_a x = \frac{1}{x \ln a}}, \qquad x > 0.$$

The natural logarithm is the simplest case because its derivative has no extra constant.

Chain rule with logarithms

If $g(x) > 0$, then

$$\boxed{\frac{d}{dx}\ln(g(x)) = \frac{g'(x)}{g(x)}.}$$

Example:

$$f(x) = \ln(x^2 + 1).$$

Since $x^2 + 1 > 0$ for every x,

$$\boxed{f'(x) = \frac{2x}{x^2 + 1}.}$$

Example: logarithm of a linear function

Let

$$f(x) = \ln(3x - 2).$$

The domain is

$$3x - 2 > 0 \quad \Longrightarrow \quad x > \frac{2}{3}.$$

Differentiating,

$$\boxed{f'(x) = \frac{3}{3x - 2}}, \qquad x > \frac{2}{3}.$$

Domain first

The derivative is meaningful only where the original function is defined.

A useful simplification before differentiating

Consider

$$f(x) = \ln \sqrt{x}, \qquad x > 0.$$

Since

$$\ln \sqrt{x} = \ln(x^{1/2}) = \frac{1}{2} \ln x,$$

we obtain

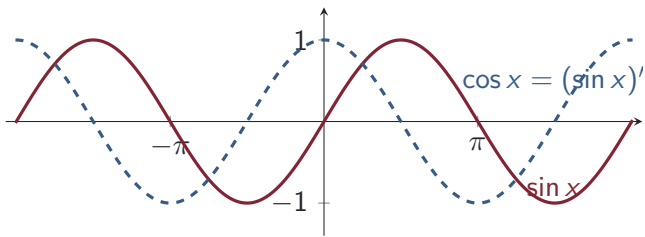
$$\boxed{f'(x) = \frac{1}{2x}.}$$

Sometimes algebraic simplification makes differentiation much easier.

Derivative of $\sin x$

Angles are measured in **radians**. Then

$$\frac{d}{dx}\sin x = \cos x.$$



Derivative of $\cos x$

Again in radians,

$$\boxed{\frac{d}{dx}\cos x = -\sin x.}$$

Thus the derivative cycle is

$$\sin x \xrightarrow{D} \cos x \xrightarrow{D} -\sin x \xrightarrow{D} -\cos x \xrightarrow{D} \sin x.$$

This cycle will be useful when studying higher-order derivatives.

Derivative of $\tan x$

Since

$$\tan x = \frac{\sin x}{\cos x},$$

the quotient rule gives

$$(\tan x)' = \frac{\cos^2 x + \sin^2 x}{\cos^2 x}.$$

Using $\sin^2 x + \cos^2 x = 1$,

$$\boxed{\frac{d}{dx}\tan x = \frac{1}{\cos^2 x} = \sec^2 x.}$$

This holds wherever $\cos x \neq 0$.

Chain rule with trigonometric functions

If $u = g(x)$,

$$\boxed{\frac{d}{dx}\sin(g(x)) = \cos(g(x))g'(x),}$$

$$\boxed{\frac{d}{dx}\cos(g(x)) = -\sin(g(x))g'(x),}$$

$$\boxed{\frac{d}{dx}\tan(g(x)) = \frac{g'(x)}{\cos^2(g(x))}.}$$

The same chain-rule structure appears again.

Examples with trigonometric compositions

$$\frac{d}{dx}\sin(3x-1) = 3\cos(3x-1).$$

$$\frac{d}{dx}\cos(x^2) = -2x\sin(x^2).$$

$$\frac{d}{dx}\tan(2x) = \frac{2}{\cos^2(2x)}.$$

Pattern

Derivative of the outer trigonometric function $\times$ derivative of the inner function.

Mixed example: product + chain rule

Differentiate

$$f(x) = e^{x^2} \sin x.$$

Product rule:

$$f'(x) = \left(e^{x^2}\right)' \sin x + e^{x^2} \cos x.$$

Chain rule:

$$\left(e^{x^2}\right)' = 2xe^{x^2}.$$

Therefore

$$\boxed{f'(x) = e^{x^2}(2x \sin x + \cos x).}$$

Mixed example: quotient + logarithm

Differentiate

$$f(x) = \frac{\ln x}{x}, \qquad x > 0.$$

By the quotient rule,

$$f'(x) = \frac{(1/x)x - \ln x}{x^2}.$$

Hence

$$\boxed{f'(x) = \frac{1 - \ln x}{x^2}}, \qquad x > 0.$$

Later the sign of this derivative will identify the maximum of $\ln x / x$.

A compact derivative table

$f(x)$	$f'(x)$
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$1/x$
$\log_a x$	$1/(x \ln a)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$1/\cos^2 x$

Logarithmic differentiation: why use it?

Some functions are awkward to differentiate directly, for example

$$y = x^x, \qquad x > 0.$$

Taking logarithms turns powers into products:

$$\ln y = x \ln x.$$

Now differentiate implicitly with respect to x.

Example: derivative of x^x

From

$$\ln y = x \ln x,$$

we obtain

$$\frac{y'}{y} = \ln x + 1.$$

Since $y = x^x$,

$$\boxed{\frac{d}{dx} x^x = x^x (\ln x + 1)}, \qquad x > 0.$$

This is neither the ordinary power rule nor the ordinary exponential rule: both base and exponent vary.

General real powers

For $x > 0$ and a real constant α,

$$x^{\alpha} = e^{\alpha \ln x}.$$

Therefore

$$\frac{d}{dx} x^{\alpha} = e^{\alpha \ln x} \frac{\alpha}{x} = \alpha x^{\alpha - 1}.$$

Thus, on its natural real domain,

$$\boxed{\frac{d}{dx} x^{\alpha} = \alpha x^{\alpha - 1}.}$$

This extends the power rule beyond positive integers.

Higher-order derivatives

If f' is differentiable, we differentiate again:

$$\boxed{f''(x) = \frac{d}{dx} f'(x).}$$

More generally,

$$f^{(n)}(x)$$

denotes the nth derivative.

Interpretation

f' measures change; f'' measures how the rate of change itself changes.

Example: successive derivatives

Let

$$f(x) = x^4 - 2x^2 + 1.$$

Then

$$f'(x) = 4x^3 - 4x,$$

$$f''(x) = 12x^2 - 4,$$

$$f'''(x) = 24x,$$

$$f^{(4)}(x) = 24, \qquad f^{(5)}(x) = 0.$$

For polynomials, sufficiently high derivatives eventually vanish.

Higher derivatives of sine and cosine

Starting from

$$f(x) = \sin x,$$

we get

$$f'(x) = \cos x,$$
$$f''(x) = -\sin x,$$
$$f'''(x) = -\cos x,$$
$$f^{(4)}(x) = \sin x.$$

The derivatives repeat with period 4.

Exercise: exponential and logarithmic derivatives

Exercise Differentiate.

1. $f(x) = e^{3x}$.
2. $g(x) = e^{x^2+1}$.
3. $h(x) = \ln(x^2 + 4)$.
4. $p(x) = \ln(5x - 1)$.
5. $q(x) = 2^x$.

State the domain whenever it is not all of $\mathbb{R}$.

Solutions: exponential and logarithmic derivatives

Solution

$$f'(x) = 3e^{3x}.$$

$$g'(x) = 2xe^{x^2+1}.$$

$$h'(x) = \frac{2x}{x^2+4}.$$

$$p'(x) = \frac{5}{5x-1}, \qquad x > \frac{1}{5}.$$

$$q'(x) = 2^x \ln 2.$$

Exercise: trigonometric derivatives

Exercise Differentiate.

1. $f(x) = \sin(2x)$.
2. $g(x) = \cos(x^2 + 1)$.
3. $h(x) = x \sin x$.
4. $p(x) = \dfrac{\sin x}{x}$, $x \neq 0$.
5. $q(x) = \tan(3x)$.

Solutions: trigonometric derivatives

Solution

$$f'(x) = 2\cos(2x).$$

$$g'(x) = -2x\sin(x^2+1).$$

$$h'(x) = \sin x + x\cos x.$$

$$p'(x) = \frac{x\cos x - \sin x}{x^2}.$$

$$q'(x) = \frac{3}{\cos^2(3x)}.$$

Mixed exercise

Exercise Differentiate

$$f(x) = e^{x^2} \ln(1 + x^2).$$

Use the product rule, then the chain rule in both factors. **Solution**

$$f'(x) = 2xe^{x^2} \ln(1 + x^2) + e^{x^2} \frac{2x}{1 + x^2}.$$

Hence

$$\boxed{f'(x) = 2xe^{x^2} \left[\ln(1 + x^2) + \frac{1}{1 + x^2} \right].}$$

Logarithmic derivative and relative change

For a positive differentiable function f,

$$\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)}.$$

Thus

$$\boxed{\frac{f'(x)}{f(x)}}$$

is the instantaneous relative rate of change. This quantity appears naturally in growth rates and elasticity calculations.

Common mistakes

- $(e^{g(x)})' \neq e^{g(x)}$ unless $g'(x) = 1$.
- $(\ln g(x))' \neq 1/g(x)$: the factor $g'(x)$ is missing.
- $(\cos g(x))'$ carries a minus sign.
- Trigonometric derivative formulas assume radians.
- Do not use the power rule on x^x.
- Always retain the domain of the original function.

Choose the outermost rule first

For

$$f(x) = \frac{e^{x^2} \sin x}{\ln(x+2)},$$

the outermost operation is a **quotient**. Inside the numerator there is a **product**. Inside e^{x^2} there is a **composition**. Inside $\ln(x+2)$ there is another **composition**.

Reliable strategy

Work from the outside structure toward the inner layers.

Concept check

Exercise Which formula is correct?

$$\frac{d}{dx}\ln(1+x^2) = ?$$

1. $\dfrac{1}{1+x^2}$.
2. $\dfrac{2x}{1+x^2}$.
3. $\dfrac{2}{1+x^2}$.
4. $2x\ln(1+x^2)$.

Solution The correct answer is

$$\boxed{\frac{2x}{1+x^2}.}$$

Complete differentiation toolbox

We now know how to differentiate combinations of

powers, products, quotients, compositions,
exponentials, logs and trigonometric functions

The next question is no longer only

"How do we compute $f'(x)$?"

but

"What does the derivative tell us about f?"

Independent practice

Exercise Differentiate and simplify when convenient.

1. $f(x) = e^{2x-1}(x^2+1)$.
2. $g(x) = \ln(1+x^4)$.
3. $h(x) = \sin(x^3)$.
4. $p(x) = \dfrac{e^x}{1+x^2}$.
5. $q(x) = x^x$ for $x > 0$.

For one function, also compute the second derivative.

Independent practice: answers

Solution

$$f'(x) = e^{2x-1}(2(x^2+1)+2x).$$

$$g'(x) = \frac{4x^3}{1+x^4}.$$

$$h'(x) = 3x^2\cos(x^3).$$

$$p'(x) = \frac{e^x(1+x^2)-2xe^x}{(1+x^2)^2}.$$

$$q'(x) = x^x(\ln x + 1).$$

Looking ahead

Now that the derivative toolbox is complete, we turn to theorems that connect local slopes with global behaviour:

$$\boxed{\text{Fermat} \longrightarrow \text{Rolle} \longrightarrow \text{Lagrange / MVT.}}$$

These results will justify our later methods for

monotonicity, extrema and complete function study.